

AWS

Solutions Architect Associate

SAA-C03 Study Notes by [Kehinde Oyewale](#) ↗

Compute • Storage • Networking • Databases • Analytics • Security • Migration • Cost • Disaster Recovery

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EC2 Pricing Models

EC2 Plan	Requirement	Discount	Commitment
On-Demand	Short-term workload. Most flexible. Priciest	None	< 1 year
Reserved Instance	Long-term predictable workload	72%	1 or 3 years
Convertible Reserved Instances	Long-term. Flexible to change EC2 reqts. EC2 only	66%	1 or 3 years
Savings Plan	Long-term commitment. Flexibility. Compute savings: EC2, Fargate, Lambda Instance savings: EC2 only. 1 family, region. Bigger discount than compute	Same as RI	1 or 3 years. \$/hr
Spot	Short workloads. Cheapest, least reliable. Stateless.	90%	None
Dedicated Host	Dedicated physical server/CPU. BYOL licensing.	Costliest	
Dedicated Instance	Dedicated/isolated hardware. No host control or sharing	Cheaper than DH	
Capacity Reservation (on-demand)	Guaranteed on-demand capacity in 1 AZ	None	None
Zonal RI	Guaranteed reserved capacity in 1 AZ (Standard or Convertible)	Same as Standard/Convertible RI	1 or 3 years
Regional RI	Across AZs in a region, no capacity, just discount		

S3 Storage Classes

S3 Class	Retrieval	Min storage duration	Use when	Cost
Standard	Millisec / instant	None	Frequently accessed, hot data	\$0.023 — costliest
Intelligent-Tiering	Millisec	None	Unknown/unpredictable access patterns	\$0.0025–0.023 — 2nd costliest
Standard-IA	Millisec	30 days	Infrequent access, needs instant retrieval	\$0.0125 — 3rd costliest
One Zone-IA	Millisec	30 days	Recreatable , infrequent access. 1 AZ only	\$0.01 — 4th costliest
Glacier Instant Retrieval	Millisec	90 days	Archive touched ~quarterly, needed instantly	\$0.004
Glacier Flexible Retrieval	Expedited 1–5 min Standard 3–5 hr Bulk 5–12 hr	90 days	Archive, minutes–hours retrieval OK	\$0.0036

S3 Class	Retrieval	Min storage duration	Use when	Cost
Glacier Deep Archive	Standard 12 hr Bulk 48 hr	180 days	Cheapest. Retrieval is 12 hr min!	\$0.00099 — cheapest

Provisioned retrieval capacity: purchase it for highly reliable Glacier expedited retrieval under any circumstance. Each unit gives 3 expedited retrievals every 5 minutes and up to 150 MB/s throughput.

Networking & VPC

VPC: allows you to choose your IP address range, create subnets, configure route tables, network gateways, securely connect AWS to on-premises.

VPC Flow Logs: for internal traffic, IP address.

On-prem ↔ VPC connections

Site-to-Site VPN: encrypted on-prem ↔ AWS over internet, IPsec, fast to set up (hours), cheap. Can be used as a failover for dedicated Direct Connect links: **DX + VPN failover**.

- **Customer Gateway (CGW):** your **on-prem** router in the VPN connection
- **Virtual Private Gateway (VGW):** AWS-side VPN endpoint, attached to 1 VPC. **IPv4 only** for Site-to-Site VPN traffic (IPv6 VPN needs a Transit Gateway)
- **VPN tunnel** runs in the middle. **2 tunnels per VPN connection** for redundancy (typically 1 carries traffic, 1 is failover)
- **Transit Gateway:** equal-cost multi-path (**ECMP**) routing, many VPCs over many VPN tunnels

Direct Connect (DX): on-prem ↔ AWS on a **private + dedicated** connection via **VIF**. Weeks–months setup, for consistent bandwidth/performance, large regular data. Not encrypted by default.

Virtual Interfaces: how you connect on-prem ↔ AWS and sort traffic on Direct Connect.

- **Private VIF:** on-prem → **1 VPC** via **Virtual Private Gateway**
- **Public VIF:** connects to **AWS services**, e.g. S3
- **Transit VIF:** on-prem → **many VPCs** via **Transit Gateway**. TGW supports **ECMP** routing over many VPN tunnels

VPN CloudHub: communicate with multiple sites over the internet using VPN, with/without VPC.

VPC ↔ VPC connections

VPC Peering: 1-to-1 VPC peering, connect 2–3 VPCs, full network access. Not transitive. **No CIDR overlap**. Cheap.

Transit Gateway: many VPCs + VPN. On-prem to many VPCs (Site-to-Site, Direct Connect). Transitive routing. Supports IP multicast (1-to-many) and many-to-many via **ECMP**.

PrivateLink: privately connect VPC to 1 **specific service** (AWS service, VPC-exposed service, SaaS partner's service). No network merge, **CIDRs can overlap**.

Internet access

NAT Gateway: sits in a **public subnet/IP**. Lets resources in a private subnet (via its route table) connect to the internet through the internet gateway. 1 NAT per AZ. Internet can't initiate on its own — outbound only.

Private subnet (route table) → NAT GW (public subnet) → Internet Gateway → INTERNET → IGW
→ NAT GW → Private subnet

Public subnet in a non-default VPC: no automatic IP is assigned. To access the internet → associate an Elastic/public IP + configure the route table + attach an IGW.

Egress-only IGW: outbound-only, IPv6 NAT equivalent.

VPC Endpoints

VPC Endpoint: private access to reach AWS services or PrivateLink.

- **Gateway Endpoint:** route-table entry. VPC connects to **S3, DynamoDB**. Free.
- **Interface Endpoint:** creates an **ENI** with a **private IP** in the subnet. For PrivateLink and other AWS services. Reachable over peering/Transit Gateway/VPN. Pay per hr/GB.

Security Groups vs NACLs

Firewall / traffic control	SG	NACL
Rules	Allow only , default deny	Allow and deny
Scope	EC2 network interface firewall	Subnet firewall
State	Stateful	Stateless
Inbound traffic	Default deny	Custom NACL: default deny Default NACL (auto-created with VPC): default allow
Outbound traffic	Default allow	Custom NACL: default deny Default NACL: default allow

Inbound access rules: **/32** CIDR = 1 IP. **/0** = all IPs.

Bastion Host: EC2 sitting in a **public subnet** / **Elastic IP for SSH/RDP** access (define this in SG) to private EC2 from the internet. Also called a **jump box**.

Session Manager: newer way to access private instances **without SSH keys or an open inbound port 22**, minimal operational overhead — just use the **SSM agent**.

ENI: virtual network card in a VPC you can attach to a new instance on failover within 1 AZ. **1 AZ**, unlike Elastic IP. Used for Lambda networking into VPCs. **Dual-homing:** attach multiple network interfaces to 1 instance. Components: private/secondary/public IPv4, Elastic IP, SG, MAC address.

Service Quotas for ENI/IP are important, esp. with Lambda. If the ENI quota limit is reached, it may fail to scale and cause errors. If Lambda can't provision the ENIs/IPs it needs: **EC2ThrottledException**.

Network Firewall: managed firewall for the whole VPC, **Layers 3–7**. Intrusion detection/prevention, domain filtering, traffic inspection/filtering, stateful rules.

Load Balancing & Resilience

NLB: 1 static IP per AZ, supports static/Elastic IP. PrivateLink support. **Layer 4**. TCP/TLS/UDP ports. Best performance. Helps *whitelist a fixed IP*. **Regional**.

- **Instance ID targets:** NLB always routes to the target's **primary private IP**.
- **IP address targets:** you can route to any IP — primary, secondary, etc.

ALB: **Layer 7**. HTTP, HTTPS, gRPC, WebSocket. Routes by path, hostname, header/query string. **Regional** — use Global Accelerator or Route 53 for cross-region.

- **SNI (Server Name Indication):** lets 1 ALB serve/host many HTTPS domains, each with its own **TLS** certificate. Many secure websites can share 1 IP address and port
- **X-Forwarded-For:** use when the *app* needs the client IP behind the ALB
- **Health checks:** instance reachability behind the ALB depends on them. Unhealthy instances are removed from rotation. An **ASG**, if configured with ELB health checks, auto-terminates and replaces them

GWLb: **Layer 3**. IP packets (GENEVE). Third-party app firewalls. Intrusion Detection/Prevention Systems.

- **Route 53 (DNS)** survives a REGION dying (DNS failover — minutes, TTL-bound)

- **ELB** survives an **INSTANCE** dying (in-path, near-instant rerouting)
- **ASG REPLACES** the dead instance (self-healing capacity)

EBS Volume Types

SSD = random small, frequent I/O operations. Boot volumes. IOPS.

- **gp3 — General purpose.** Good price/performance, cheap. Baseline **3,000 IOPS / 125 MiB/s** at any size; scales to **80,000 IOPS / 2,000 MiB/s, 64 TiB** (previously 16K IOPS — older exam material may still say 16K). IOPS independent of size. **gp2** is the older version (IOPS tied to size, 3 IOPS/GB).
- **io1 / io2** — Critical business apps, Provisioned IOPS, sustained IOPS performance. **io1**: up to **64,000 IOPS**, 50 IOPS/GB — for I/O-intensive workloads. **io2 Block Express** (all io2 volumes are Block Express since April 2025): up to **256,000 IOPS, 4,000 MB/s, 64 TiB, 1,000 IOPS/GB**, sub-millisecond latency, 99.999% durability — highest performance, lowest latency, same price as io1. **Multi-Attach**: up to 16 Nitro instances in 1 AZ can attach the same volume.

HDD = can't be boot volumes.

- **st1**: Throughput Optimized. Frequent access. Big, sequential data. Log processing. Cheap.
- **sc1**: Cold archive. Infrequent access. Cheapest.
- **Magnetic volumes**: even cheaper, for infrequent access. Legacy.

Category	Types	Boot volume?
SSD	gp2, gp3, io1, io2	Yes
HDD	st1, sc1, Magnetic (legacy)	No

EC2 Instance Types

- **Storage optimized (I, D)**: high, sequential read/write access to big data. Low-latency, high random IOPS. OLTP systems, SQL/NoSQL on local disk, data warehouses, distributed systems.
- **Compute optimized (C)**: high CPU. Batch processing, media transcoding, HPC, ML, game servers, scientific modeling.
- **Memory optimized (R, X)**: RAM. In-memory cache, real-time big data analytics. **X** — extreme RAM, large scale, SAP HANA. SQL/NoSQL DBs, BI.
- **General purpose (M, T)**: balanced CPU/memory/network. **T** — spiky, idle, burstable. **M** — general, fixed performance.
- **Accelerated computing (G, P, Inf)**: **G** — GPU graphics. **P** — power graphics, ML training. **Inf** — ML inference (AWS Inferentia chips).

Amazon Machine Image (AMI): customization (OS, software, config) of EC2. The image EC2 uses to spin up a new instance.

Launch template: the template an ASG uses to launch EC2. You specify instance info, e.g. AMI ID, instance type, key pair, SGs, block device mapping. Can't be modified — create a new version instead.

Nitro: newer EC2 hypervisor/hardware platform — higher performance and throughput ceilings, especially relevant to EBS IOPS limits (e.g. io2 Block Express 256K, io1 64K need Nitro).

Instance & File Storage

EBS: **private, persistent** personal network drive; persists even after instance termination (**DeleteOnTermination**: root volume default true, extra data volumes default false). Supports live changes — usable while creating a snapshot, while attached, while detached and reattached, changing volume type/size, reads/writes. **1 AZ**, 1 instance except io1/io2 Multi-

Attach. Boot volumes/DBs. Provisioned capacity. **Encrypted** snapshots can't be public. KMS (symmetric keys) for access, **KMS + IAM policy** to share.

EFS: shared EC2 storage — multi-AZ, multi-instance, **Linux**, POSIX, **NFS**, serverless, autoscales, high IOPS. Pricier than EBS.

Instance Store: fast but disposable, not persistent. Cache/buffer.

Data Lifecycle Manager (DLM): automates EBS backups or snapshot creation, retention, deletion on schedule. EBS snapshots are stored in S3.

FSx for Windows File Server: managed Windows file server — **SMB, NTFS**, Active Directory integration, Windows file shares.

FSx for Lustre: **HPC, Linux + cluster, massively parallel, high-throughput**, sub-millisecond latency, 100s GB/s throughput, works with S3. ML training, video rendering, genomics, financial modeling.

FSx for NetApp ONTAP: multi-protocol support — **NFS, SMB, block iSCSI**. Most flexible. **Windows, Linux, Mac. Storage efficiency:** deduplication, compression, thin provisioning. **SnapMirror:** continuous data replication across regions. On-prem NetApp → AWS migration.

FSx for OpenZFS: snapshots, cloning, compression, high IOPS, sub-millisecond latency. Big data analytics, **NAS** migration, media processing.

Storage Gateway (Hybrid)

Storage Gateway: live, **ongoing** hybrid storage that apps write to/store in — not just migration. NFS, SMB (File Gateway), iSCSI (Volume Gateway).

S3 File Gateway: **NFS or SMB** file share access to S3 for on-prem applications, with a **local cache** for low-latency access to recent data.

FSx File Gateway: low-latency on-prem access to **FSx for Windows** shares with a local cache. **No longer available to new customers since Oct 2024** — may still appear in older exam questions.

Volume Gateway: **block storage** via **iSCSI** volumes. Modes:

- **Cached volumes:** primary data lives in S3, frequently accessed data cached on-prem.
- **Stored volumes:** primary data **on-prem**, low-latency local access, **async backup to S3 as EBS snapshots**.

Tape Gateway: replaces **physical tape backup infrastructure** with a **virtual tape library (VTL)**.

Scenario triggers

- "On-prem + S3 + SMB/NFS + cache/low latency" → S3 File Gateway
- "Windows file server + SMB + AD/NTFS + Windows applications" → FSx for Windows File Server

Protocol quick map

Protocol	Services
SMB / NFS	S3 File Gateway, FSx for ONTAP
SMB / NTFS	FSx for Windows File Server
NFS only	EFS
Block / iSCSI	Volume Gateway, FSx for ONTAP
Block (EC2-attached)	EBS

IAM Policy Types

- **Identity-based policies:** attach to user/group/role. AWS managed policy, inline policy (single user/role), customer managed policy (self-managed).
- **Resource-based policies:** grant permissions to a specified principal; attach to a resource: S3 bucket policy (ACLs are legacy), KMS, Lambda, SQS/SNS and **IAM role trust policy**.
- **IAM role trust policy:** resource-based policy for **cross-account access**. Specify your trusted principal (AWS account/resource, federated IdP); they call `sts:AssumeRole` for temporary credentials.
- **Permissions boundaries:** attach to a **user or role**; set the max permissions identity-based policies can grant an entity. Doesn't grant permissions itself.
- **SCP:** policies for **organizational units/accounts** to restrict users and roles/permissions.
- **AWS Resource Access Manager (RAM): cross-account sharing** — centrally share resources (e.g. Transit Gateway) with other AWS accounts/OUs.
- **VPC endpoint policies:** control VPC principals and resources, for gateway and interface endpoints.

IAM/resource policies: explicit deny wins. **NACL:** lowest rule number wins.

RDS

RDS storage auto scales — you set/provision maximum storage. PITR (automated backup retention up to 35 days). Continuous backups. Multi-AZ standby **synchronous** setup for AZ failure/DR. Up to 15 **async** read replicas within AZ, cross-AZ or cross-region. **Encryption can't be enabled after creation** — snapshot, copy with encryption enabled, then restore.

- **Database Activity Streams:** near-real-time audit feature that captures and streams DB activity events outside the database for monitoring and compliance. Destination: **Kinesis Data Streams**.
- **RDS Proxy: Lambda + RDS** connection pool that fixes the "**too many DB connections**" / **connection exhaustion** error, allowing connections to scale to 1000s.
- **RDS Custom / EC2:** custom DB configuration and OS-level access. RDS Custom = managed **Oracle / SQL Server** only.
- **Enhanced Monitoring:** granular OS-level data — OS processes, RDS child processes, load average.
- **RDS event subscriptions:** near-real-time event notifications via **SNS (or EventBridge)** on instance start/stop, failovers, backups, config changes, deletion, maintenance events. Invoke **Lambda functions** directly for data-level events.

RDS Multi-AZ Deployments	RDS Read Replicas (up to 15)
Synchronous replication – highly durable	Asynchronous replication – highly scalable
Only DB on primary instance is active	All read replicas accessible, used for read scaling
Automated backups taken from standby	No backups by default
Automatic failover to standby for DR (flip CNAME)	Manually promoted to a standalone DB instance
Span two AZs within a single Region	Can be within an AZ, cross-AZ, or cross-Region
DB version upgrades happen on primary	DB upgrade independent from source instance

Aurora

Aurora: live app transactions, OLTP. Autogrows to 128 TB, no provisioning. **3 AZs, 6 copies (2 per AZ), 15 read replicas**. Self-heals. 3–5x faster than RDS.

- **Failover — single instance:** create a new primary instance in the same AZ; for AZ failure, create it in a **different AZ**. **Automatic**.

- **Failover — with replicas:** flip the cluster endpoint (its CNAME) to point to a healthy replica. **Promote replica to new primary/writer.** ~30 secs. **Automatic.**

Aurora Global Database: secondary **cross-region replication for active-passive failover / DR.** **Manual failover.** 1 primary region (up to 15 read replicas), up to 10 secondary regions (up to 16 read replicas each). RPO ~1 sec, RTO under 1 min. Low latency.

Aurora Serverless: autoscales by usage, pay per use. **Unpredictable/spiky/idle/intermittent work**, e.g. dev DB.

Automated backups — 35 days PITR (same as DynamoDB/RDS). **Manual snapshots — forever.** **1 Aurora Capacity Unit (ACU) ≈ 2 GB memory.**

Aurora Endpoints:

- **Writer / cluster endpoint** → **writes** → primary instance. 1 writer.
- **Reader endpoint** → **reads** → up to 15 read replicas. Load balances. 1 reader endpoint per cluster.
- **Instance endpoint** → targets 1 specific DB instance in a cluster.
- **Custom endpoint** → you define and group your own specific set of instances. Load balances.

DynamoDB

DynamoDB Streams: CRUD modifications. Real-time, async. 24 hrs retention. Uses Lambda triggers or the DynamoDB Streams Kinesis Adapter.

DynamoDB Global Tables: **multi-region active-active** DR. Apps can READ/WRITE to the tables in any region simultaneously, all active, low latency. DynamoDB Streams must be enabled.

DynamoDB Auto Scaling: dynamically adjusts provisioned throughput capacity (including GSIs' RCU/WCU) to handle sudden increases in traffic without throttling.

Adaptive Capacity: DynamoDB auto-shifts throughput capacity when partitions aren't evenly distributed.

DynamoDB Accelerator (DAX): in-memory cache. **Microsecond** reads, API-compatible, zero code change.

GSI: different partition + sort key, create anytime, deletable. Independent provisioned capacity (RCUs/WCUs).

LSI: same partition key / different sort key, table creation time only, can't delete, shares base table RCU/WCU. Never use for unknown data size / unpredictable growth — 10 GB item collection limit. Can't add for a new query pattern in prod. Strongly consistent reads.

S3 Features

S3: objects up to **50 TB** (via multipart upload). **Single PUT 5 GB max. Single GET 5 TB max** — use concurrent/byte-range GETs for bigger objects. Data is assigned an object key used for retrieval. Strong read-after-write consistency.

- **S3 Transfer Acceleration** → global **UPLOAD** speedup to 1 S3 bucket. Routes upload via CloudFront edge to reduce distance/latency.
- **Multi-Region Access Points:** single unified global endpoint for closest / lowest-latency bucket routing based on requester location.
- **S3 Access Points:** **dedicated access policy** per team, optional **VPC restriction**.
- **S3 Cross-Region Replication:** only replicates new objects by default. **Enable versioning** first. For compliance, DR, low latency. Near real time.
- **S3 Batch Replication:** replicates existing objects.
- **S3 Lifecycle rules:** expiration actions, transition actions. Can delete or move after a period.
- **Object Lock:** **WORM** — prevents overwrite and delete. Requires **versioning**. **Compliance mode:** nobody can overwrite/delete, not even root. **Governance mode:** admins with special permission can. **Legal hold:** nobody, indefinitely, until lifted — **no retention period**.
- **Multipart upload** → split a large upload into chunks. Recommended if data > 100 MB.

- **Byte-range fetch** → download part of an object, e.g. header/footer of a huge file.
- **S3 Bucket Key**: fixes heavy traffic hitting **KMS API request limits**, cutting **KMS API costs by up to 99%**. Trade-off: coarser per-object audit trail in CloudTrail.
- **Directory bucket**: stores objects in S3 Express One Zone. For low latency.
- **Table bucket**: Apache Iceberg format. For ML analytics, tabular data. Athena, Redshift, Spark.
- **S3 Batch Operations**: perform bulk operations on existing S3 objects with a single request.

S3 encryption

Option	Who manages the key	Notes
SSE-S3	S3/AWS auto-manages keys and rotation	AES-256 . Default for new objects. Simplest
SSE-KMS (header aws : kms)	KMS-managed	Key usage audit logs in CloudTrail , access control via key policy
SSE-KMS — AWS managed key	KMS auto-manages the key, rotates yearly	No control over rotation or key policy
SSE-KMS — customer managed key	You create/manage the key in KMS	You choose whether to enable rotation, control key policy, can disable/delete
SSE-C (customer-provided keys)	You supply the key with every request; S3 encrypts/decrypts, AWS never stores the key	HTTPS required. You handle rotation
Client-side encryption	You encrypt before upload , you manage keys and rotation	AWS never sees plaintext

S3 caveats

- S3 CRR/SRR only replicate new objects by default; to replicate existing objects, use **S3 Batch Replication**.
- **Versioning must be enabled** for Cross-Region Replication and Object Lock.
- S3 event notification has **one destination per event rule**, so apply SNS fan-out to multiple SQS queues/other services.
- S3 event notification destinations accept Standard SNS, never SNS/SQS FIFO. Free.
- Event notifications only fire on new events, not for existing objects.
- **Supported event destinations: SNS, SQS, Lambda, EventBridge.**

SQS / SNS

SQS: decouple, buffer, queue. **SNS**: pub/sub messaging/notifications.

- SNS FIFO topics can **ONLY** fan out to SQS FIFO queues.
- **SNS Standard topic destinations**: Lambda, standard SQS, email/SMS, HTTP endpoints (plus Firehose, mobile push).
- SQS queue retention — 4 days default, increase to 14 days max with the SetQueueAttributes action.
- SQS visibility timeout — 30 seconds default, max 12 hours. If it expires before the consumer processes/deletes the message, the message becomes visible again.
- **SQS FIFO** — 300 msg/sec without batching, 3,000 msg/sec with batching. Standard SQS is near-unlimited.

S3 event → Standard SNS only. **SNS FIFO** → **SQS FIFO** only.

Lambda & Edge Compute

Lambda: 15 min max execution time. Default concurrency limit of 1,000 — request a quota increase or configure concurrency to scale and prevent **throttling**. Scales within seconds, unlike ASG minutes. Pay per use, no expensive

overprovisioning. Ephemeral storage 512 MB default, up to 10 GB. **Deployment package limit: 50 MB zipped, 250 MB unzipped** — use **container images** for larger (up to 10 GB).

- **Synchronous:** directly trigger Lambda → API Gateway/ALB waits for response. Real-time processing.
- **Asynchronous invocation:** Lambda responds to events pushed to it, queues them and **retries (2x max)** on failure, then sends to a **Dead Letter Queue/destination**. Invoked by **S3 event notifications, SNS, EventBridge**.
- **Event source mappings (polling):** Lambda reads items from other streams/queues — DynamoDB, SQS, Kinesis, Kafka, MQ, MSK.
- **Provisioned Concurrency:** pre-warmed. Fixes cold-start latency, milliseconds. **You pay for it always**, even when idle.
- **SnapStart:** cached snapshot. Fixes cold-start latency at no extra cost. Java (also Python and .NET).
- **Reserved Concurrency:** sets a cap to limit concurrency and prevent overwhelming downstream systems; also guarantees capacity for the function. No extra charge.
- **Lambda authorizer:** custom Lambda function that checks whether an API Gateway request is authorized.

CloudFront Functions: JavaScript, lightweight. Must execute in milliseconds; cheaper and faster than Lambda@Edge. Higher scale per second. Viewer Request and Viewer Response only. 10 KB size.

- **Viewer Request:** modify headers/cookies, normalize cache keys, query strings, URL redirects/rewrites, auth checks, generate HTTP responses.
- **Viewer Response:** insert security headers, modify response codes.

Lambda@Edge: Viewer Request/Response **and** Origin Request/Response. Runs Lambda code/capabilities at CloudFront edge. Node.js/Python. Higher latency, 5–10 secs execution time. < 10 MB size. Can access network, filesystem. Can modify HTTP request/response body. For adjustable CPU/memory, third-party libraries, AWS SDK.

API Gateway & Step Functions

API Gateway: REST API, HTTP API (proxy, JWT, cheap), WebSocket API. Throttles on traffic spikes — **429 Too Many Requests**. 29 sec timeout. *"Offer customers different API request limits per pricing tier"* → **Usage Plans + API Keys (REST)**. Accepts/processes 100,000s of concurrent API calls, including traffic management, authorization/access control, monitoring, API version management.

Step Functions: helps with API Gateway and Lambda timeouts; coordinates sequenced and **parallel** steps with built-in *retries, error handling, long-running workflows*.

CloudFront & Global Accelerator

CloudFront: CDN. **Caches** content at the edge — **static assets** (images, videos, JS/CSS, downloadable files, API responses) and **cacheable dynamic content**. Layer 7 (HTTP/HTTPS). **Signed URL: access ONE file. Signed Cookies: MANY files**. Brings CONTENT closer (at the edge).

- **Origin failover:** auto-switch from primary to secondary in **origin groups** on failure.
- **Origin Access Control:** makes the S3 bucket private and grants CloudFront alone the right to read it.

Global Accelerator: routes traffic to the closest edge location via Anycast — 2 static anycast IPs (fixed IPs for whitelisting). Makes the JOURNEY/traffic faster. **TCP/UDP**, Layer 4 (gaming, IoT, VoIP). Dynamic/real-time. No caching. Instant failover.

- **CloudFront** → global DOWNLOAD/view of content (HTTP)
- **Global Accelerator** → global TCP/UDP apps
- **S3 Transfer Acceleration** → global UPLOAD to S3 via CloudFront edge

Route 53

- **Public hosted zone:** answers queries from the whole internet.
- **Private hosted zone:** answers queries **only from inside VPCs you attach it to**.

- **Multivalued answer routing:** returns up to **8 healthy IP addresses** for a single DNS query, automatically excludes unhealthy ones, works across regions.
- **Geoproximity routing:** route based on the location of users and resources, with a **bias** to shift more or less traffic to a resource.

Active-active failover: when you want all your resources available the majority of the time. All records with the same name/type (A or AAAA) and same routing policy (**weighted or latency**) are active unless Route 53 considers them unhealthy.

Active-passive failover: primary resource available most of the time, secondary resource on standby. To configure:

- **1 primary and 1 secondary resource:** use Failover routing policy + health check.
- **Multiple primary and secondary resources:** Failover routing + Weighted routing set to primary. Give each record the same name, type, routing policy.

DNS Records

- **A:** hostname → IPv4 (literal, fixed IP)
- **AAAA:** hostname → IPv6 (literal, fixed IP)
- **CNAME:** hostname → another hostname (DNS name that has an A/AAAA IP), e.g. ELB DNS, CDN, S3 website endpoint. **Illegal at root/apex.** Queries cost money.
- **Alias:** hostname → another AWS resource's hostname (ALB, CDN, S3 endpoint). CAN exist at the root/apex. Free.
- Alias, A, AAAA can point at the root apex — not CNAME.
- Alias/CNAME can point to changeable DNS (CDN/ALB/S3) — not A/AAAA (fixed IPs; CDN/ALB IPs change).
- **Root domain + AWS = Alias.**

EventBridge

EventBridge: serverless event router with intelligent filtering and multi-destination routing. Native target for **Step Functions** and **Kinesis Data Streams**.

- **Rules: rich content filtering** for event buses, e.g. route only events where a specific JSON field matches a value, to different targets.
- **Event Buses:** receive/capture incoming events and route them, e.g. to Lambda/SNS. 1-to-many targets.
- **EventBridge Pipes:** simplify integrations between producers and targets. 1 source → 1 target.
- **Scheduled rules / cron** with Lambda.
- **Archive/replay events.**
- **Schema registry.**

Monitoring & Auditing

CloudWatch: performance/health metrics, logs, alarms (can terminate, stop, reboot, recover an instance). **Container**

Insights: monitor EKS and ECS metrics/logs.

- **Custom metrics:** memory, disk space/swap, logs, page file. Use the **CloudWatch agent**. Mnemonic: **DSSM-PLC**
- **Default metrics:** CPU, network, and disk read/write ops. Mnemonic: **RWNDA-CPU**

CloudTrail = API audit. Free 90-day event history. For longer: **Trail → S3 + log file integrity validation.** **CloudTrail Lake:** SQL querying.

- **Management events:** account/admin-level actions — create/terminate/modify resources.
- **Data events:** data-level actions — S3 **object** ops, Lambda invokes. Disabled by default.
- **Log file integrity validation:** proves CloudTrail log files haven't been modified, deleted or tampered with (SHA-256 digest files).
- **CloudTrail Insights:** detects **unusual API activity** (e.g. sudden spike in TerminateInstances).

Config: view and record configuration changes on a timeline. SNS notifications for alerts.

- **Config rules:** audit, evaluate **compliance**; **EventBridge** for notifications on **non-compliance**.
- **SSM remediation:** fix non-compliant resources with **SSM Automation documents**, retries possible. (To *prevent* changes, use SCP/IAM.)

Analytics & Big Data

Redshift: managed **petabyte-scale, columnar data warehouse** for **complex analytical queries** across huge datasets. **OLAP**, PostgreSQL-based, Massively Parallel Processing (MPP). Two modes: provisioned cluster and Serverless. 10x performance. SQL. BI tools, e.g. QuickSight or Tableau. Faster than **Athena** (OLAP, S3 query).

- **Leader node:** query planning, results aggregation.
- **Compute node:** performs queries, sends results to leader.

Redshift Spectrum: lets Redshift reach **out** to query S3 directly.

Glue: batch ETL. Data Catalog. Serverless. Convert data to Parquet/ORC. For static data, e.g. S3.

- **Glue Job Bookmarks:** prevent re-processing old data.
- **Glue DataBrew:** clean and normalize data using pre-built transformations.
- **Glue Studio:** GUI to create, run and monitor ETL jobs in Glue.
- **Glue Streaming ETL:** process continuous data streams in near real time.

EMR: big-data processing/analytics/migration for Hadoop/Spark/Hive clusters, ML, BI. Serverless option available. Used with On-Demand, Reserved, Spot.

- **Master node:** delegates and monitors.
- **Core node:** stores data and processes tasks.
- **Task node:** optional extra processing — use **Spot**.

Lake Formation: automates/simplifies **setting up a centralized organization data lake** for analytics and ML. **Granular** permissions. Built on top of Glue. Cleanse, transform, ingest data. S3, RDS, NoSQL.

Kinesis: big data streaming/analytics (**ETL**). Log/clickstream analytics, live dashboards/leaderboards, metric generation and reporting, sensor data, IoT, ad tech. Enhanced fan-out.

Kinesis Data Streams: big data ingestion/storage (up to 1 year; 24 hrs default). MULTIPLE consumers read the SAME data. Can replay/reprocess. Data ordered in shards, sequenced. Provisioned or On-Demand (pay per use). Can't delete stream data. Metrics and reporting. Real-time. **Kinesis Producer Library (KPL):** continuous writes. **Kinesis Client Library (KCL):** read/query, ordered distributed processing.

Kinesis Data Firehose: **loads** stream data into data lakes/storage/analytics, e.g. S3, Redshift, OpenSearch, custom HTTP endpoint. **Basic transform** with Lambda, e.g. convert JSON/CSV to Parquet/ORC. Auto scaling, serverless, pay per use. Managed/no code. Near real time — 60 sec buffer, unlike KDS milliseconds.

Kinesis Data Analytics / Managed Service for Apache Flink: real-time **ETL** processing/SQL on a **live stream**.

Kinesis Video Streams: ingest and process real-time video streams.

Quick picks

- **ETL:** Glue (batch), KDA (real-time), Firehose (light transform + load only)
- **Data analytics:** KDA, Redshift, EMR
- **S3 query:** Redshift Spectrum, Athena
- **Storage:** Redshift (OLAP), Lake Formation (centralized, granular), KDS (real-time streams, ordered)

Service	Role
S3	Data lake (raw storage)
Redshift	OLAP data warehouse (analytics + storage)

Service	Role
Redshift Spectrum	Query S3 from Redshift
Athena	Serverless query engine for S3
Glue	Batch ETL , prepares data for S3/Redshift, includes Data Catalog
Kinesis Data Streams	Temporary live streaming ingestion pipe
Kinesis Data Analytics	Real-time ETL processing/SQL on the stream
Kinesis Firehose	Delivers stream data to S3/Redshift/etc. (with optional light transform)
Lake Formation	Centralized governance/security ON TOP of an S3 data lake, granular access control, simplified lake setup
EMR	Big data analysis — Hive, Spark, Hadoop

	Kinesis Data Streams	DynamoDB Streams
Consumers / retention	Many consumers. 1 year max, 24 hrs default	Limited consumers. 24 hours
Process with	Lambda, Kinesis Data Analytics, Data Firehose, Glue Streaming ETL	Lambda, DynamoDB Streams Kinesis Adapter

Containers

EKS: open-source, cloud-agnostic orchestrator. Managed node (EC2) groups, self-managed nodes, Fargate. **aws-auth ConfigMap** maps IAM users/roles to K8s RBAC via AWS IAM Authenticator for K8s.

- **Cluster Autoscaler:** older, cloud-agnostic, works at ASG/node-group level, simple setup, self-managed.
- **Karpenter:** faster, more lightweight/flexible, cheaper, self-managed autoscaler.
- **EKS Auto Mode:** fully AWS-managed autoscaling, built on Karpenter.
- **IRSA (IAM Roles for Service Accounts):** fine-grained IAM permissions for individual Pods in an **EKS** cluster, leveraging an OpenID Connect (OIDC) identity provider.

ECS: container platform/orchestrator. For IAM: ECS task roles (defined in task definitions). **EC2 + ASG** to scale yourself, **or capacity providers + Managed Scaling** for autoscaling.

Role / policy	What it controls
ECS Task Execution Role	Permissions ECS needs before launch / before app code runs (pull image from ECR, write logs, fetch secrets)
ECS Task Role	What the container CAN DO at runtime
Lambda Execution Role	What Lambda CAN DO when it needs AWS permissions
Lambda Resource-Based Policy	WHO can invoke it — specify the principal (account/service)

EC2 launch type: you provision and maintain instances **with the ECS agent** (runs on EC2), unlike the **Fargate launch type**. Supports GPUs.

Fargate: serverless compute engine (EC2 isn't serverless). Just create task definitions. **No GPU**. Works with ECS and EKS.

ECR: stores container images, image vulnerability scans.

ECS/EKS: orchestrators. **Fargate/EC2:** compute engines (EC2/Fargate launch types).

ECS Anywhere: run ECS-managed containers on your own on-prem/non-AWS infrastructure.

EKS Anywhere: EKS Distro + AWS support/tooling; run Kubernetes on your own (customer-managed) infrastructure.

EKS Distro: the **open-source** Kubernetes software EKS uses underneath, for running K8s yourself anywhere.

Wavelength: edge of 5G telecom networks, ultra-low latency.

Local Zones: low latency for large population centers; an extension of a single parent region (not multi-region).

Migration & Data Transfer

DataSync: accelerated bulk **data/file migration** or transfer from on-prem to AWS. **One-time or scheduled**. Fast, replaces manual scripts. Needs good internet (or Direct Connect). Needs the **DataSync agent unless it's AWS-to-AWS**. Moves between S3 ↔ EFS ↔ FSx ↔ **NFS** ↔ **SMB** shares.

Database Migration Service (DMS): migrate databases to AWS with **near-zero downtime**, with the **Schema Conversion Tool**. **Continuous** data replication (**Change Data Capture**) — ongoing streaming. Sources/targets include SQL/NoSQL, S3, Redshift, Redis, Kafka. Supports heterogeneous migration.

AWS Transfer Family: managed **SFTP/FTPS/FTP** server/endpoint for file transfers in/out of **S3/EFS**.

Data Transfer Terminal: physical facility where you bring your storage devices to quickly transfer data in minutes. No network **bandwidth** needed.

Snow Family:

- **Snowcone:** fast, tiny.
- **Snowball Edge Compute Optimized:** edge processing, limited bandwidth.
- **Snowball Edge Storage Optimized:** 50–500 TB.
- **Snowmobile:** 10–100 PB shipping-container truck.
- **Snowball can't write directly to Glacier** — land in S3, then use a lifecycle policy.

Application Migration Service (MGN) / AWS Transform: **lift-and-shift**. Migrate hundreds of physical/virtual **servers** to EC2 with **minimal downtime**, continuous automated block-level replication. 90 days free.

Elastic Disaster Recovery: MGN's twin (continuous automated block-level replication) but for **disaster recovery**. **Cheap DR** for physical/virtual/cloud servers. RPO: seconds. RTO: 5–20 mins.

Application Discovery Service: plan cloud migration by collecting usage and configuration data about on-prem servers.

Migration Hub: aggregates migration info for visibility.

Quick picks: Minimal downtime → DMS, MGN. Low bandwidth → Data Transfer Terminal, Snowball.

Application Integration

AppSync: GraphQL real-time API for **web/mobile apps** to combine data from multiple backends/microservices. Real-time queries, mutations, subscriptions. DynamoDB, Lambda, RDS, HTTP.

AppFlow: sync/transfer data from external SaaS apps (Salesforce, Slack, etc.) into S3/Redshift without custom code.

Amazon MQ: managed message broker service (JMS, AMQP) for **migrating legacy applications already built on traditional message brokers** (ActiveMQ/RabbitMQ).

Identity & Directory Services

Cognito User Pools: **authentication** (sign-up, sign-in, MFA, password reset) for **end users**. Issues ID, access and refresh tokens (JWT). Supports social/federated IdPs, e.g. Google, Facebook and SAML.

Cognito Identity Pools: **authorization** — lets end users access AWS services directly. Exchanges a User Pool JWT (or social/SAML identity) for **temporary AWS credentials via STS**.

IAM Identity Center: grants an **organization's staff** access to AWS via **SSO**. Centralized workforce identity and access control, **permission sets** for granular access. Supports AD, the default **Identity Center directory**, SAML, third-party/external IdPs (e.g. Okta). Temporary AWS credentials via STS.

Managed Microsoft AD: a real, standalone AD **living in AWS** (new directory). Create a **two-way forest trust relationship** to connect it to on-prem AD for on-prem and AWS Identity Center benefits.

AD Connector: **proxy** between AWS and **existing on-prem AD**. Forwards auth requests to on-prem AD. Connects WorkSpaces and AWS console sign-in to on-prem AD.

Simple AD: lightweight, cheaper, smaller-scale **Samba-based** directory.

WorkSpaces: cloud-hosted **virtual desktop (VDI)** service — users access a full Windows or Linux desktop from any device. Pairs with AD.

Directory scenario triggers

- On-prem AD ↔ AWS Managed Microsoft AD → two-way forest trust
- On-prem AD used directly by AWS services → AD Connector
- Users need Windows desktops with Microsoft AD authentication → WorkSpaces + Managed Microsoft AD
- Employees need SSO to multiple AWS accounts → IAM Identity Center
- Existing on-prem AD, AWS services should use it → AD Connector

Security Services

CloudHSM: dedicated, single-tenant hardware security modules in your VPC. **FIPS 140-2 Level 3** validated (US govt crypto standard). Strict **regulation/compliance**. You manage your keys.

Control Tower: set up/govern a multi-account environment with governance, compliance, guardrails. **Account Factory:** self-service account creation with pre-approved configs. **Preventive guardrails:** SCPs. **Detective guardrails:** Config rules. Account drift notifications, logging, audit, SSO.

GuardDuty: threat detection. Continuously monitors for **malicious/unauthorized activity** with ML. Escalate to **Detective**. Sources: CloudTrail logs, VPC Flow Logs, DNS logs, EKS audit logs, S3 data events.

Inspector: vulnerability scans on EC2, container images (ECR), Lambda.

Macie: ML-powered PII / **sensitive data discovery** in S3.

Artifact: compliance report downloads. **Security Hub:** aggregation dashboard.

WAF (Web Application Firewall): protects against common Layer 7 web exploits — XSS, SQL injection, geo-match (block countries), rate-based rules for DDoS.

Firewall Manager: centralized WAF, Shield and SG policy management across an Organization.

Shield: automatic DDoS protection, Layer 3/4. **Shield Advanced:** paid, AWS DDoS experts, Layer 3–7.

Secrets Manager: automatic secret rotation (DB credentials, API keys). **KMS:** key encryption. **Parameter Store:** cheap/free config store.

KMS envelope encryption / GenerateDataKey: encrypt a large file with KMS. **Level 2.**

AWS Certificate Manager (ACM)

ACM: create, store, and renew public/private TLS certs. Free auto-renewal. Integrates with ELB/CloudFront/API Gateway. Wildcard certs protect unlimited subdomains. Regional.

ACM Private CA: build a public key infrastructure (PKI) inside AWS for private use within an organization (VPN, microservices, laptops, etc.).

- The private key of a **public ACM certificate can't be exported**. If you need the cert outside AWS, use an external CA (e.g. Let's Encrypt) + **IAM certificate store**, or better still:

- **Private CA certs** and their private keys **can be exported** for your PKI outside AWS.
- Certificates for **CloudFront MUST live in us-east-1** (any region works for a regional ALB).
- You can't add/remove domains on an existing certificate — request a **new** cert. Can't delete a certificate in use either.
- To use with EC2, use a load balancer (with the public cert) or export a Private/third-party CA cert.
- TLS/SSL encrypts in-flight data — use **ACM**. **KMS** encrypts data at rest.
- To renew an external/imported certificate, obtain a new cert from your issuer, then manually reimport it into ACM.

Compute, Caching & Databases

Elastic Fabric Adapter (EFA): network interface for EC2 apps with high levels of inter-node communication at scale; accelerates HPC/ML workloads (performance, latency, bandwidth). Essentially an **ENA (higher PPS) with OS bypass** — communicates directly with the network interface. OS bypass is Linux only, not Windows.

Elastic Beanstalk: PaaS. Deploy/upload code; AWS auto-handles provisioning, load balancing, scaling, but you retain control over the resources. Stores logs in S3, CloudWatch.

ElastiCache Redis: data durability. HA (Multi-AZ, read replicas). **Sorted sets** for real-time gaming leaderboards.

ElastiCache Memcached: simple, multithreaded, non-persistent. Multi-node partitioning/sharding. Auto Discovery identifies cluster nodes.

Neptune: managed graph DB (relationships between entities, e.g. social networks). **Neptune Streams:** real-time ordered sequence of changes. HTTP REST API.

DocumentDB: managed, MongoDB-compatible document database.

Keyspaces: managed, Cassandra-compatible wide-column NoSQL database.

Placement Groups

- Specify the number of instances you need at launch and use 1 instance type, else you might get an insufficient capacity error. Stop and restart the instances if it happens.
- You can't merge placement groups. An instance can only be in 1 placement group at a time.
- **Dedicated Hosts not supported.** Spot Instances configured to **stop or hibernate** on interruption not supported.

Cluster placement group	Spread placement group	Partition placement group
Many instances per rack. No rack limit (1 AZ, packed close together)	Many racks (7/AZ), 1 instance each	Many partitions (7/AZ), each with many racks and many instances
Lowest latency / HPC / tightly coupled	Few critical instances, isolate failures. SOLO	Hadoop / Cassandra / Kafka at scale. Big distributed systems. Packed
No 7-per-AZ cap	7 instances per AZ	7 partitions per AZ

Disaster Recovery Strategies

Strategy	What's running normally	RTO	Cost
Backup & Restore	Nothing (just backups in S3). For DR, provision everything from scratch with the backups: spin up new instances, restore DB from a snapshot	Hours	Cheapest
Pilot Light	Only the most critical core (e.g. DB) at minimal size in DR. Everything else exists as AMIs/templates ready to launch	Tens of minutes	Low
Warm Standby	Everything, scaled down (fewer/smaller instances) but FULLY FUNCTIONAL	Minutes	Medium

Strategy	What's running normally	RTO	Cost
Multi-Site Active-Active	Everything, full scale	Near-zero	Most expensive

CloudFormation

CloudFormation: infrastructure as code. Model, provision and version-control your entire infra in a text file.

- **Change Sets:** **preview** what an update would create/modify/delete *before* you apply it.
- **Drift Detection:** detect manual changes to resources.
- **Stack Policies:** protect specific resources in a stack **from being modified by stack updates**.
- **StackSets:** deploy **one template to MANY accounts and regions** in one operation. *"Roll out a security baseline across the organization."*
- **Nested Stacks:** stacks inside stacks; package common patterns (e.g. a standard VPC) as reusable components.
- **DeletionPolicy:** per-resource instruction on stack deletion — **Retain** or **Snapshot** (RDS, EBS, etc.).
- **Instance Scheduler on AWS:** an AWS Solution deployed via a CloudFormation template — automatically starts/stops EC2 and RDS instances on a schedule to cut costs (up to 70% for non-24/7 workloads).

Cost Management

Tool	One-liner
Cost Explorer	Visualize, analyze, filter historical spend, forecast ~12 months ahead, get RI/Savings Plan recommendations
AWS Budgets	Set cost or usage thresholds, alert/notify when exceeded
Cost & Usage Report (CUR)	Most detailed, granular billing data, delivered to S3 , queried with Athena (or QuickSight). SQL
Cost Anomaly Detection	ML watches for unusual spikes/anomalies in spend and alerts you
Compute Optimizer	ML rightsizing recommendations for EC2 instance size, EBS volumes, Lambda, ASG
Cost Allocation Tags	Tag resources per team/project, then slice costs by tag. Activate in the Billing console

Operations & Management

AWS Systems Manager (SSM): operate AWS/on-prem infra at scale without SSH/RDP access.

- **Session Manager:** secure shell access without opening inbound ports or managing SSH keys.
- **Run Command:** execute commands or scripts **remotely, across many instances simultaneously**, without logging into each one.
- **Patch Manager:** automates **OS patching** across an EC2 fleet.
- **State Manager:** ensures **instance state/configuration**; continuously checks and corrects drift.
- **Automation:** custom maintenance workflows, triggerable via **EventBridge**.
- **Inventory:** collects instance metadata (installed applications, OS versions, network configs) for visibility and compliance audits.
- **Fleet Manager:** console to view/manage groups of instances.
- **Parameter Store:** cheap/free config and env store. **SecureString** parameters are encrypted with KMS. No built-in rotation (use Secrets Manager for that).

AWS Backup: centralized backups across multiple resources — schedules, retention, etc.

Batch: queued batch or scheduled jobs. **ParallelCluster:** traditional HPC cluster management.

Health Dashboard: personalized view of AWS service health/issues affecting your account.

License Manager: track and manage software licenses (plus BYOL) across AWS resources.

Service Catalog: admins curate pre-approved, configured stack **resources** for teams to self-serve.

Well-Architected Tool: self-service review tool; checks architecture against AWS's best-practice pillars.

Managed Grafana: managed dashboards/visualization for metrics.

Managed Prometheus: managed metrics collection/monitoring, Prometheus-compatible.

Other Services

Device Farm: test mobile apps on real physical devices in the cloud.

Amplify: simplified frontend-to-backend app development/deployment toolkit.

Serverless Application Repository: catalog of pre-built, deployable serverless app templates.

VMware Cloud on AWS: run your existing VMware stack on dedicated AWS hardware, no re-architecting.

Elastic Transcoder: converts media files between formats (older, superseded by MediaConvert).

AI/ML Services

- **Rekognition:** face detection, labeling, celebrity recognition
- **Transcribe:** speech to text, e.g. subtitles, voice recordings
- **Polly:** text to speech
- **Translate:** translations
- **Lex:** build conversational bots — chatbots
- **Connect:** cloud contact center
- **Comprehend:** natural language processing (NLP). Extract phrases, places, people; find insights
- **Comprehend Medical:** detects information in unstructured clinical text. Detects Protected Health Info (PHI) — DetectPHI API
- **SageMaker:** ML for developers and data scientists
- **Kendra:** ML-powered search engine
- **Personalize:** real-time personalized recommendations
- **Textract:** detect text and data in documents

CIDR Cheat Sheet

CIDR	Total IPs $2^{(32 - \text{CIDR})}$	Usable (- 5)
/32	1	— (single host)
/28	16	11
/27	32	27
/26	64	59
/25	128	123
/24	256	251
/23	512	507

CIDR	Total IPs $2^{(32 - \text{CIDR})}$	Usable (- 5)
/22	1,024	1,019
/16	65,536	65,531

Common Ports

Type	Protocol	Port	Traffic
SSH	TCP	22	Encrypted
HTTP	TCP	80	Unencrypted
HTTPS	TCP	443	Encrypted (TLS/SSL)
RDP	TCP / UDP	3389	Encrypted
FTP	TCP	21	Unencrypted
SFTP	TCP	22 (runs over SSH)	Encrypted

ARN Format

`arn:aws:service:region:account-id:resource-type/resource-id`

Example: `arn:aws:ds:us-east-1:123456789012:directory/d-1234567890`